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LEVEL 3 · LEARNER MANUAL

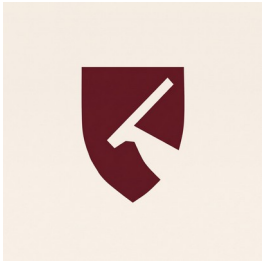
Advanced Safety Management

Safety. Preparedness. Competence.

Author: Tangi Iigonda

Oryx Institute · Oryx Safety Academy

Ten modular days · Eight modules · Certificate of Completion



Advanced Safety Management — Learner Manual

Oryx Safety Academy · Oryx Institute

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Welcome from the Principal

Welcome to Level 3. You arrive with authority, or about to receive it: a department, a group of sites, a budget line with your name next to it. This programme assumes that authority and teaches the crafts that keep it honest, systems that survive your absence, investigations that change organisations rather than blame individuals, audits that find what the polished centre hides, and training that multiplies rather than merely informs.

Ten days, five modules, taken together or in pairs. The teaching is Namibian and the labels are honest: what is law, what is an adopted benchmark, and what must still be verified with an authority before it is written into your documents. Hold us to that standard in the room, and to every other one in the programme.



Tangi Iigonda, Principal and Founder, Oryx Institute

About this programme

Field	Detail
Duration	Ten days, delivered as five two-day modules; modules may be taken separately
Audience	Senior safety officers, SHE and SHEQ managers, compliance managers and consultants serving Namibian industry.
Entry requirement	Level 2 Safety Representative and Site Safety Officer, or demonstrated equivalent responsibility and training. Learners taking single modules need the relevant prerequisite experience, assessed on enrolment.
Assessment	One written examination per module (8 questions, 75% pass) plus two practical assignments: the management-system assignment (Module 1) and the practical audit report (Module 7). Rubrics issued with the assessment pack.

Field	Detail
Certificate	Certificate of Completion — Advanced Safety Management (internal credential; not an NQF qualification)
Modularity	Modules 1–8 may be taken separately; the Certificate of Completion requires all eight module examinations and both assignments within eighteen months.

Certification status, stated plainly: the Oryx Safety Academy is not yet a registered vocational training provider with the Namibia Training Authority, and this certificate is not an NQF qualification. The Training-of-Trainers module certifies teaching craft; it does not substitute for the safety credential required of lead instructors, nor for statutory mine appointments, nor for certification in first aid or fire, which run through registered partners. Formal vocational programmes commence after NTA provider registration; NQA accreditation is being pursued in parallel and will be announced on the public register.

The ten days

Days	Module
Days 1–2	Module 1 — SHEQ management systems and ISO 45001 principles (gap-analysis assignment set)
Day 3	Module 2 — Construction safety requirements in depth (method-statement surgery)
Day 4	Module 3 — Mining safety requirements (code-of-practice skeleton)
Day 5	Module 4 — Occupational hygiene and environmental monitoring (survey commissioning)
Day 6	Module 5 — Contractor management (prequalification panel)
Days 7–8	Module 6 — Root-cause investigation: ICAM fundamentals (full investigation with board panel)
Day 9	Module 7 — Auditing and inspection techniques (audit exercise)
Day 10	Module 8 — Training of trainers (assessed training segments); module examinations; close

How to use this manual: each module carries outcomes, terminology, teaching content, a worked example, an activity and a knowledge check. Module examinations draw from the taught material and its applications. The manual is designed to remain your desk reference long after the programme ends.

Module 1 SHEQ management systems and ISO 45001 principles

Contact time: Two days · *Oryx Safety Academy*

1.1 What this module covers

This module builds the management-system view of safety: policy, planning, support, operation, evaluation and improvement wired together so the system, not heroics, produces performance. ISO 45001:2018 provides the reference architecture; the module teaches its logic, not certification consultancy, and how to run a gap analysis against it.

1.2 Learning outcomes

By the end of this module you should be able to:

- Explain the PDCA cycle and map the ISO 45001:2018 clause structure onto it.
- Describe what each major clause family demands, context, leadership, planning, support, operation, evaluation, improvement.
- Identify the legal-and-other-requirements inputs a Namibian organisation must feed the system, and how to keep them current.
- Conduct a structured gap analysis of an organisation against ISO 45001 principles and prioritise the gaps.
- Design the documentation architecture that serves the system without burying it.

1.3 Key terminology

Term	Working definition
SHEQ	Safety, health, environment and quality managed as one integrated system.
PDCA	Plan-Do-Check-Act, the operating cycle of management systems.
Context of the organisation	The internal and external issues, and interested parties, that shape the system's scope.
Gap analysis	A structured comparison of current practice against a standard's requirements, prioritised for action.

Term	Working definition
Documented information	The standard's term for anything the organisation must control as documents or records.

1.4 Why systems outperform campaigns

Campaigns spike attention and decay; systems persist through staff turnover, contract changes and management fashion. The difference is architecture: a management system defines who owns risk, how competence is built, how performance is measured, and how the machine corrects itself when drift appears. A site with a working system answers questions in minutes that a site without one answers with meetings.

ISO 45001:2018 is the international reference for such systems. It is voluntary in Namibia; clients increasingly write it into contracts, and certification is a commercial decision separate from the safety benefit. The standard's value for a manager is its checklist completeness: it is hard to forget a whole clause family.

1.5 The clause map on one page

Context and scope: the issues and parties that define the system's boundaries. Leadership: top management's visible commitment, the policy, roles and consultation with workers. Planning: hazard identification, legal and other requirements, objectives and the programmes that resource them. Support: resources, competence, awareness, communication, documented information. Operation: operational control, managing change, procurement including contractors, emergency preparedness, and readiness to respond. Performance evaluation: monitoring, measurement, analysis, evaluation, internal audit, management review. Improvement: incident handling, nonconformity and corrective action, and continual improvement.

Run that list against a real organisation and the gaps announce themselves. The most common Namibian gaps are worker consultation that exists only on paper, management reviews that review nothing measurable, and change management that is triggered only by disaster.

1.6 Legal inputs and their maintenance

The planning clause requires a live register of legal and other requirements: the Labour Act and GN 156, licences and their conditions, client standards, and the voluntary commitments the organisation has signed. Each entry needs an owner and a review rhythm; the pending OSH Bill belongs on the register now, flagged as pending, so its enactment triggers a review instead of a scramble. The Oryx

Safety Academy Reference Register modelled in this programme is exactly this instrument at programme scale.

1.7 Gap analysis as the first project

The practical entry point is a gap analysis: clause by clause, evidence or its absence, ranked by risk and effort, converted into a twelve-month programme with owners and budgets. The assignment in this course is precisely that exercise on a supplied organisation profile, and the audit module later in the programme reuses the same discipline from the verifier's side of the table.

1.7 Worked example

Worked example: A 200-person logistics group gap-analyses against ISO 45001 principles over two weeks. Findings, ranked: consultation of drivers nonexistent (leadership/planning); no legal register beyond the Labour Act, licences untracked (planning); competence records scattered across three depots (support); subcontracted maintenance outside the system entirely (operation); management review is an annual lunch (evaluation). The programme drafted: driver safety committee with two elected reps per depot; legal register with owner and quarterly review; single competence register; contractor prequalification incorporated into procurement terms; management review agenda with six mandatory measurable inputs. Two years later the group's client audits improved from procedural passes to substantive ones, and the driver committee had produced the near-miss programme that management could not have designed from an office.

1.8 Activity and knowledge check

LEARNER ACTIVITY

Gap analysis on the supplied profile

Groups receive the Oryx logistics-group profile, forty pages of description, org chart, existing documents. Produce a clause-by-clause gap table with evidence, risk-rank the top five gaps, and draft the twelve-month programme for the top two. The class challenges each programme for ownership, resourcing and measurement.

KNOWLEDGE CHECK

SHEQ management systems and ISO 45001 principles — check your understanding

1. Map the seven ISO 45001 clause families onto PDCA.
2. Why does a system outperform a campaign?
3. What belongs on the legal-and-other-requirements register besides statutes?
4. Name the three most common gaps found in Namibian gap analyses in this module's experience.
5. Certification to ISO 45001 in Namibia is…
   A. A legal requirement
   B. A voluntary, commercial decision
   C. Automatic after one audit

1.9 Module summary

- Systems beat campaigns through architecture: ownership, competence, measurement, correction.
- ISO 45001's clause families map onto PDCA.
- Legal inputs live on a register with owners and review rhythms.
- Gap analysis, ranked and resourced, is the entry project.

Module 2 Construction safety requirements in depth

Contact time: One day · *Oryx Safety Academy*

2.1 What this module covers

Beyond the Level 2 foundation, this module treats construction as a managed production system: design-stage risk reduction, method statements, temporary works, lifting operations, occupation and handover. The Namibian legal frame remains the Labour Act and GN 156; South African instruments appear as contractual good practice, labelled honestly.

2.2 Learning outcomes

By the end of this module you should be able to:

- Apply prevention-through-design thinking to a construction programme.
- Specify the contents and discipline of method statements for high-risk activities.

- Manage temporary works, scaffolds, formwork and excavation support, as engineered items with checks.
- Plan lifting operations: equipment certification, lift categories, exclusion and supervision.
- Run a construction-phase safety programme: inductions, inspections, permits, coordination and handover.

2.3 Key terminology

Term	Working definition
Method statement	A written, task-specific safe system of work: sequence, hazards, controls, competencies, checks.
Temporary works	Engineered temporary structures, scaffolds, formwork, shoring, that carry load during construction.
Lifting plan	The document defining a lifting operation's equipment, configuration, load, ground, exclusion and supervision.
Prevention through design	Removing construction hazards on the drawing board rather than managing them on site.

2.4 Design decisions are safety decisions

The cheapest place to remove a hazard is the drawing. Permanent edge protection designed into a roof beats scaffolding and harnesses forever after; a service corridor planned above ground beats a trench through live services; prefabrication in a controlled yard beats assembly at height. The safety manager's input at design review is a series of questions: what will this detail force a worker's body to do at height, in a hole, or under a load, and what would the drawing look like if it forced nothing?

2.5 Method statements that survive contact with site

A method statement states the sequence, the hazards at each step, the controls, the competencies required, and the hold points where a named person signs before work continues. The discipline is authorship by the people who will do the work, reviewed by a competent person, and revision whenever the sequence changes. Sites recognise the difference instantly: statements written for the file get read never; statements written with the crew get corrected with a pencil, which is the system working.

2.6 Temporary works and lifting

Temporary works are structures: designed or selected by a competent person, checked before first load, and inspected on a rhythm, the scaffold-tag discipline from Level 2 scaled up to formwork and shoring. Lifting operations are planned events: the equipment certified and inspected, the lift categorised simple, standard or complex; the ground assessed; the load known; the exclusion zone marked; the signal person competent; and weather limits stated in advance. The lifting plan is written for anything beyond routine lifts, and the crane's next examination date is on the plan.

2.7 The construction-phase programme

Construction-phase safety is a rhythm: induction for every person before their first task; daily high-risk briefings; weekly inspections with reports; permits for the fatal-minute activities; coordination meetings where trades sequence their hazards past each other; and handover documentation that transfers knowledge instead of liabilities. The safety file from Level 2 is the spine of the rhythm; this module is about making it pulse.

2.7 Worked example

Worked example: A warehouse build in Otavi adopts prevention through design after its safety manager joins the design review: roof-mounted plant is relocated to a ground-level compound, removing 300 hours of work at height; a precast stair core replaces temporary ladders between levels; services are rerouted around the crane pad identified as sitting above a stormwater culvert. Construction-phase costs rise slightly at design and fall heavily at site; the injury forecast disappears from the height and excavation categories entirely. The lifting plan for the remaining steel is one page: crane certificate numbers, the complex-lift categorisation, exclusion diagram, wind limit, and the appointed person's name.

2.8 Activity and knowledge check

LEARNER ACTIVITY

Method statement surgery

Groups receive a real-format method statement for excavation and pipe-laying, deliberately weakened: vague sequence, no hold points, competencies missing, trench support unspecified. Repair it: full sequence, hazards per step, controls, competencies, hold points with named roles. Swap groups and audit the repaired version against a checklist: could a new foreman run the job from this document

alone?

KNOWLEDGE CHECK

Construction safety requirements in depth — check your understanding

1. Why is the drawing the cheapest place to remove a hazard?
2. List the five elements of a method statement that survives site contact.
3. Temporary works must be…

   A. Built to look strong

   B. Designed/selected by a competent person, checked before first load, inspected on a rhythm

   C. Inspected only if they fail
4. What appears on a written lifting plan?
5. Name the five rhythms of construction-phase safety.

2.9 Module summary

- Design review is a safety tool; ask what the drawing forces bodies to do.
- Method statements are written with the crew and revised with the sequence.
- Temporary works and lifts are engineered, checked events.
- The construction phase runs on rhythm; the safety file is its spine.

Module 3 Mining safety requirements

Contact time: One day · *Oryx Safety Academy*

3.1 What this module covers

Mining concentrates hazards at industrial scale: ground, mobile equipment, energy, dust and isolation. This module teaches transferable mine-site safety management while being scrupulously honest about the legal position: Namibia's mining industry operates under the Minerals (Prospecting and Mining) Act, 1992, administered by the Ministry of Mines and Energy, with mine safety requirements applied through that framework, while many operations use the South African Mine Health and Safety Act as the regional benchmark. Current

Namibian mine-specific safety regulations must be verified with the Ministry before being cited in any document.

3.2 Learning outcomes

By the end of this module you should be able to:

- Describe the Namibian regulatory position for mining, including what must be verified before citation.
- Manage the classic mine hazard families: ground control, mobile equipment interaction, energy isolation, dust and ventilation basics.
- Structure the elements of a mine safety programme: risk assessment, codes of practice, training, and health surveillance.
- Explain why codes of practice and shift-start meetings dominate mine safety communication.

3.3 Key terminology

Term	Working definition
Code of practice (mine)	A site-specific documented procedure for managing a identified major hazard, a regional mining convention.
Ground control	The management of rock mass stability, support and seismics in underground and open-pit operations.
Energy isolation	Lock-out and tag-out of electrical, hydraulic, pneumatic and gravitational energy before work.
Health surveillance	Scheduled medical monitoring of workers exposed to hazards such as noise and dust.

3.4 The legal position, stated honestly

The Minerals (Prospecting and Mining) Act, 1992 (Act 33 of 1992) governs Namibian mining through the Ministry of Mines and Energy. Mine safety requirements operate within that framework and any licensing conditions attached to operations. The South African Mine Health and Safety Act, 1996 (Act 29 of 1996), with its regulations, is widely applied across southern African operations as the benchmark and is frequently written into Namibian clients' and operators' standards. The professional rule this course teaches: in any Namibian mining document, cite Namibian law as law, cite the MHSA or SANS instruments as adopted benchmarks, and mark mine-specific Namibian regulatory citations as “to be verified with the Ministry of Mines and Energy” until confirmed in writing. This course's own reference register carries that flag on every mining entry.

3.5 The hazard families

Ground: support standards installed within tolerance, inspected at shift start, with fall-of-ground reporting treated as the mine's heartbeat. Mobile equipment: segregation of people and plant by design, visibility management, and interaction rules at haulage crossings and tips. Energy: lock-out with personal locks and tags before any work on powered or pressurised systems. Dust and ventilation: airflow engineered and monitored, respirable dust controlled at source with wet methods and extraction, and exposure verified by measurement rather than assumption. Isolation of working alone: nobody works alone in unventilated or unsupported ground, ever, and the buddy rule is enforced by the absence of exceptions.

3.6 The programme elements

A mine safety programme runs on: site-wide and task risk assessments; codes of practice for each major hazard, ground, vehicles, explosives, ventilation; induction and annual re-induction; shift-start meetings that carry yesterday's learnings; statutory-type inspections recorded; and health surveillance with exposure records retained for decades, because occupational disease latency is measured in decades. The safety manager's craft is the same as anywhere else in this programme: hazards named, controls resourced, checks recorded, learning loop closed. Only the scale and the consequence of drift are larger.

3.7 Worked example

Worked example: An open-pit operation near Karibib introduces a simple interaction rule after two near-misses at the pit ramp: every light vehicle carries a flashing beacon, headlights on, radio channel logged, and gives way to haul trucks at marked intersections; trucks announce on radio at each ramp approach. Cost: beacons, radios, signage, one afternoon of induction. Within a quarter, the near-miss category “vehicle interaction” drops to zero reports, which the safety manager correctly treats as a signal to verify reporting is still alive, an old lesson: improving a metric can hide the death of the measurement that produced it.

3.8 Activity and knowledge check

LEARNER ACTIVITY

Build the code of practice skeleton

Groups draft the skeleton of a code of practice for one assigned hazard, mobile equipment interaction or fall of ground: purpose, scope, definitions, hazard description, control standards, roles, training requirements, inspection and records,

review trigger. Swap and mark: which section, if left vague, would first cause an incident?

KNOWLEDGE CHECK

Mining safety requirements — check your understanding

1. What is the correct citation practice for Namibian mining safety documents, as taught here?
2. Name the four classic mine hazard families in this module.
3. Why are health surveillance records kept for decades in mining?
4. Fall-of-ground reporting is treated as the mine's heartbeat because…
   A. It feeds production statistics
   B. It is the earliest signal of ground-control drift before a collapse
   C. Regulations demand it alone
5. A safety metric improves to zero. What is the professional response?

3.9 Module summary

- Cite Namibian mining law as law, benchmarks as benchmarks; verify mine-specific regulations.
- Ground, vehicles, energy, dust: the families and their disciplines.
- Codes of practice, shift-start meetings, health surveillance: the programme spine.
- Zero can mean fixed or unmeasured; check which.

Module 4 Occupational hygiene and environmental monitoring

Contact time: One day · *Oryx Safety Academy*

4.1 What this module covers

Occupational hygiene is the science of measuring what you cannot see: dust, noise, heat, vapours and fumes at concentrations that injure slowly. This module teaches anticipation, recognition, evaluation and control of health hazards, survey design, the meaning of exposure limits, and how to commission and read monitoring reports without pretending to be a hygienist.

4.2 Learning outcomes

By the end of this module you should be able to:

- Run the hygiene cycle: anticipate, recognise, evaluate, control, re-verify.
- Design a basic exposure survey: who, where, when, how long, what method.
- Interpret exposure results against occupational exposure limits, including the statistical treatment of shift exposure.
- Apply the control hierarchy to health hazards, with ventilation and process change leading.
- Commission specialist monitoring competently and audit the report's quality.

4.3 Key terminology

Term	Working definition
Occupational exposure limit (OEL)	An airborne concentration ceiling used to judge exposure acceptability; source-dependent and jurisdiction-dependent.
Personal sampling	Measurement in a worker's breathing zone, over a representative shift.
8-hour time-weighted average (TWA)	Exposure averaged over a workday, the standard comparison basis for OELs.
Baseline survey	The first systematic measurement establishing exposure status for a process or group.

4.4 The cycle and the honesty of measurement

Anticipate from process knowledge: what does this process liberate? Recognise by walkthrough and worker interview: where does it hurt, itch, smell or settle? Evaluate by measurement designed to answer a question, not to decorate a file: personal sampling in breathing zones, over full shifts, across the seasons that change the exposure. Control with the hierarchy, ventilation and process change before personal protection, and re-verify afterwards, because a control that is installed is not a control that is working.

The Namibian practice point: many Namibian employers apply South African OEL tables, from the mining and general-industry frameworks, or international lists such as the ACGIH threshold limit values, as benchmarks in the absence of an exhaustive Namibian schedule. That is workable good practice; the

professional requirement is to name the source of every limit used, in every report, and never to present a benchmark as a Namibian statutory limit.

4.5 Reading a monitoring report

A competent report states: what was sampled, by which method and laboratory; how many personal samples over how many shifts; the results against named limits; and the interpretation. The manager's reading questions: are the samples personal or static, a static sample by a machine describes the machine, not the worker; is the TWA arithmetic shown for the longer shifts common in Namibian operations, a 12-hour shift can double a TWA that looked compliant at 8 hours; and what does the survey recommend, in the hierarchy's order, with dates. Reports without those elements are decoration.

4.6 The classic Namibian exposures

Dust: silica in quarrying, brick cutting, mining and construction, controlled by wet methods, extraction and enclosed cabs, with respirator programmes as the last line and health surveillance for the decades. Noise: plant rooms, grinders, haulage, with buy-quiet procurement and enclosure before hearing protection, audiometry as surveillance. Heat: outdoor work in the Namibian summer, with the administrative controls from Level 1 and acclimatisation management. Vapours and fumes: solvents in cleaning and paints, welding fume in fabrication, controlled by substitution, ventilation and, last, respiratory protection with fit testing.

4.7 Worked example

Worked example: A bottling plant commissions a noise survey after two machine operators request audiometry. The report: 5 static samples, 2 personal samples, results 82 to 91 dB(A), limit cited as 85 dB(A) with no source named. The SHE manager's reading: static samples dominate and describe the machines; the personal samples at 88 and 89 dB(A) are the finding; the limit's source is missing and is requested; the recommendation “issue ear plugs” ignores the encoder room, where a ¥200 gasket and an enclosure would drop the exposure to below the limit for everyone. The re-survey, redesigned with 12 personal samples across both shifts and limits sourced, produces a prioritised engineering list, with hearing protection as the documented interim control, not the solution.

4.8 Activity and knowledge check

LEARNER ACTIVITY

Commission the survey

Groups write the commissioning brief for a silica exposure survey at a brick-cutting yard: population and shifts, sampling strategy, methods and analysis, OELs to be applied with sources, reporting requirements including TWA treatment for 10-hour shifts, and the hierarchy-ordered recommendations expected. Swap briefs and mark: which missing sentence would most likely produce a useless survey?

KNOWLEDGE CHECK

Occupational hygiene and environmental monitoring — check your understanding

1. Name the hygiene cycle stages.
2. What distinguishes a personal sample from a static one?
3. A 12-hour shift raises measured exposure averages because…
4. What must every OEL citation carry?
   A. Nothing; limits are universal
   B. Its source, since limits differ by jurisdiction and list
   C. The laboratory's approval
5. Why is hearing protection the last control for machine noise?

4.9 Module summary

- Anticipate, recognise, evaluate, control, re-verify.
- Personal samples, full shifts, named limits, shown TWA arithmetic.
- Benchmarks must be labelled with their source, never passed as Namibian statute.
- Engineering first; protection is interim, documented with dates.

Module 5 Contractor management

Contact time: One day · *Oryx Safety Academy*

5.1 What this module covers

Most major incidents on client sites involve contractors, not employees. This module builds the contractor lifecycle: prequalification, contract terms, induction, supervision, permits and performance review, with the client's duty to

coordinate rather than delegate away. The pervasive failure is treating contractors as someone else's problem; the module's thesis is the opposite.

5.2 Learning outcomes

By the end of this module you should be able to:

- Structure a contractor lifecycle from prequalification to post-contract review.
- Write safety terms into contracts that are enforceable and auditable.
- Design an induction that transfers the site's real risks, not a video.
- Supervise contractor work: verification versus inspection, and the permit interface.
- Run performance reviews that change the next contract.

5.3 Key terminology

Term	Working definition
Prequalification	Screening a contractor's safety capability before award, on evidence rather than promises.
Client coordination duty	The client's obligation to manage interfaces between its own operations and contractor work, which cannot be contracted away.
Verification	Checking that the contractor's stated systems actually operate on the ground.
Post-contract review	The scored record of a contractor's safety performance that feeds the next prequalification.

5.4 Prequalification on evidence

Ask for documents that can be verified: safety policy with a named accountable person; training and competence records sampled against actual site workers; incident history with explanations, a contractor with zero incidents and zero explanations is reporting nothing; equipment certification; and references from comparable work. Score it, weight it in the award decision, and record why, so that the cheapest bid cannot silently erase the safest one.

5.5 Contract terms that bite

The contract states: the site rules the contractor must meet and the standards that apply, labelled by status, law, client standard, good practice; the right to

stop dangerous work without commercial penalty; permit and reporting interfaces; insurance and compensation responsibilities; and the consequences of safety failure, from re-briefing to removal. A safety annex nobody audits is decoration; the same annex audited on day one of the works is a system.

5.6 Induction, supervision, permits

Induction transfers the site's live risks: yesterday's near misses, this month's hot-work zones, the current laydown changes, the site's people by name and role. Supervision is verification: does the method statement on paper match the crew's actual sequence; are the competent persons named actually on site; are permits closed at day's end. The client's own supervisors need the training to recognise contractor excellence and contractor theatre alike, and the standing to act on the difference.

5.7 Worked example

Worked example: A plant turnaround hires twelve contractors. The client's safety manager runs the lifecycle: prequalification scores one contractor out for fabricated competence records, a sample phone call to a named rigger's referee did not check out; the contract annex carries the stop-work right and the reporting interface; a shared daily 07:10 coordination meeting sequences the day's hot work and lifts so trades do not create hazards for each other; permits are issued at one point, by one trained person; and the post-contract review scores every contractor, feeding the next turnaround's shortlist. Two stop-work interventions during the turnaround are recorded neutrally, resolved, and cost nobody their contract, which is the point: the right to stop is only real when using it is safe.

5.8 Activity and knowledge check

LEARNER ACTIVITY

The prequalification panel

Groups receive three contractor dossiers, one strong, one theatrical, one thin. Score them against the evidence checklist, decide award, and draft the five contract safety clauses you would insist on for the chosen one. The class hears the “theatrical” dossier's reveal: what looked strongest was the emptiest. Debrief on which checks caught it.

KNOWLEDGE CHECK

Contractor management — check your understanding

1. What does prequalification screen for, and on what basis?
2. Name five safety terms a contractor contract should carry.
3. Why does induction carry yesterday's near misses?
4. Verification differs from inspection because…
5. What makes a stop-work right real?

5.9 Module summary

- Lifecycle, not paperwork: prequalify, contract, induct, supervise, review.
- Evidence beats promises; zero incidents with zero explanations means zero reporting.
- Coordination duties cannot be delegated away.
- Stop-work rights are real only when safe to use.

Module 6 Root-cause investigation: ICAM fundamentals

Contact time: Two days · *Oryx Safety Academy*

6.1 What this module covers

Level 2 taught five-whys. This module teaches structured causal analysis at organisational level, built on the ICAM (Incident Cause Analysis Method) approach derived from James Reason's organisational-accident model: preconditions, individual actions, task and environmental factors, and organisational factors behind them. The aim is investigations whose recommendations change systems, not memories.

6.2 Learning outcomes

By the end of this module you should be able to:

- Explain the organisational-accident model: latent conditions, active failures, defences.
- Structure an ICAM-style investigation: fact gathering, timeline, analysis across the four factor groups.

- Distinguish immediate causes from underlying organisational causes in a real incident.
- Write findings and recommendations that a board can resource and an auditor can verify.
- Recognise the investigation failure modes: stopping at blame, stopping at the first cause, stopping at paperwork.

6.3 Key terminology

Term	Working definition
Latent condition	A system weakness planted in the organisation long before an incident: an unenforced rule, a training gap, a budget choice.
Active failure	The human act or omission that triggers the event.
Defence	A barrier meant to stop the hazard reaching people: guards, permits, briefings, alarms.
ICAM	Incident Cause Analysis Method; an investigation framework organised around organisational factors, task and environmental factors, individual actions, and preconditions.

6.4 The organisational accident

Serious incidents rarely begin on the day they happen. Reason's model, which ICAM operationalises, pictures an organisation carrying latent conditions, a maintenance backlog, a supervisor vacancy, a schedule written by sales, until an active failure, an error anyone could make, walks through the holes in the defences. The investigation's job is to find the latent conditions, because blaming the active failure and re-training the person changes nothing the next tired worker will meet. The first question of a good investigation is not “who” but “what made this act reasonable to this person at this moment”.

6.5 Structure of the investigation

Gathering: scene, documents, and interviews, taken while fresh, separated, and without leading. Timeline: the sequence, with times, ending at the moment of harm and continuing through response. Analysis across four factor groups: organisational factors, policies, resourcing, culture; task and environmental factors, tools, time, weather, design; individual actions, errors and violations, described without character judgement; and preconditions, fatigue, training, workload, communication. Each causal line ends at the organisational level or is

marked as a genuine individual act, which is rarer than investigations claim and never an excuse to stop asking.

6.6 Findings a board can act on

A finding states: the causal factor, the evidence for it, and the system it lives in. A recommendation states: the change, the owner, the date, the cost class, and how completion will be verified. “Remind operators to be careful” fails every test; “reallocate the third shift's maintenance window so the defect list is worked before the Monday run, owned by the plant manager, by the end of Q1, verified by the internal audit's sampling” passes all of them. The investigation report ends with a corrective-action tracker, alive until every action closes.

6.7 The three failure modes

Stopping at blame: the investigation ends when someone is named. Stopping at the first cause: the first plausible explanation ends the asking. Stopping at paperwork: recommendations are accepted by management and never resourced. Each feels like completion; each guarantees the incident's return with a different name on it. The quality review of any investigation asks one question: which of the three did this report narrowly avoid?

6.7 Worked example

Worked example: Night shift, a bagging plant: an operator reaches into a running conveyor to clear a jam and loses two fingers. First analysis: operator error, retrain, discipline. ICAM analysis: task factors, the guard's interlock had nuisance-tripped for six weeks and been defeated with a cable tie kept on the machine for the purpose; organisational factors, maintenance response times had no standard, and the defect list had no owner since February; preconditions, the night shift ran one operator short for a month, making the stop-clear-restart cycle the only way to hold output; individual action, the reach-in, was the reasonable act of a person inside an unreasonable system. Recommendations: interlock redesign and nuisance-trip fix, engineering, dated; defect-list ownership and maintenance response standard, organisational, dated; staffing norm restored, resourcing, dated; the cable-tie culture named at the next toolbox talk, without naming the operator. Two years on, the plant's near-miss reporting tripled, because the investigation's honesty had been noticed.

6.8 Activity and knowledge check

LEARNER ACTIVITY

The full investigation

Groups receive a scenario pack: documents, scene photographs, and six interview transcripts for a serious forklift-pedestrian collision. Produce: the timeline, the four-factor analysis with evidence citations, three organisational-level findings, and five recommendations that pass the owner-date-verification test. Present to a “board panel” of the other groups, who must try to fund one and reject one, with reasons. Rejections that survive argument are rewrites, not defeats.

KNOWLEDGE CHECK

Root-cause investigation: ICAM fundamentals — check your understanding

1. Define latent condition and active failure with one example each.
2. Name ICAM's four factor groups.
3. What makes a recommendation board-ready?
4. State the three investigation failure modes.
5. The first question of a good investigation is…
   A. Who was involved?
   B. What made this act reasonable to this person at this moment?
   C. What did it cost?

6.9 Module summary

- Incidents grow from latent conditions meeting active failures through failed defences.
- Analyse across organisational, task/environmental, individual and precondition factors.
- Recommendations carry owners, dates, costs and verification.
- Avoid the three failure modes; review reports against them.

Module 7 Auditing and inspection techniques

Contact time: One day · *Oryx Safety Academy*

7.1 What this module covers

This module moves from walking a site to auditing a system: planning an audit programme, building checklists from standards, gathering evidence through

documents, interviews and observation, grading findings, and writing reports that get actioned. The same discipline serves internal audits, client audits and preparing to be audited.

7.2 Learning outcomes

By the end of this module you should be able to:

- Distinguish inspection, audit and verification, and schedule each correctly.
- Build an audit checklist from a standard or procedure by converting each requirement into an evidence question.
- Gather and triangulate evidence: documents, interviews, observation.
- Grade findings, major, minor, observation, improvement opportunity, and defend the grading.
- Write audit findings that meet the evidence-test and track actions to closure.

7.3 Key terminology

Term	Working definition
Audit	A systematic, documented verification against a standard or requirement.
Objective evidence	Information that can be re-verified: a record, an observation, a demonstrable fact.
Finding	A statement of nonconformity with the requirement it fails and the evidence for it.
Audit trail	The chain from requirement to evidence to finding, reconstructable by a third party.

7.4 From requirement to evidence question

Every clause of a standard or step of a procedure converts into a question with an evidence answer. “The organisation shall determine competence requirements” becomes “show me the competence requirements for rigger, and the records of the two riggers on site”. The checklist is the conversion done in advance, with columns for evidence, sample and finding. Auditors who skip the conversion interview in generalities and leave with impressions; auditors who do it leave with findings.

7.5 Triangulation and sampling

Documents say what should have happened; interviews say what people believe happens; observation shows what happens. Any two in conflict is a finding in itself. Sample deliberately: newest worker, oldest worker, night shift, the record everyone is proud of and the record nobody offered. Audit the system's edge cases, because the centre is always polished.

7.6 Grading and writing

A finding is written as three sentences: the requirement, the evidence, the gap. Grade by risk, not by irritation: major, the system fails at a point that could allow serious harm or a statutory breach; minor, an isolated lapse in a working system; observation, a weakness that invites drift; improvement opportunity, beyond compliance, worth doing. Every finding needs a root-cause answer from the auditee and an action with owner and date, tracked to closure like any corrective action. The auditor's discipline is courtesy: audits that humiliate produce sites that hide.

7.7 Worked example

Worked example: An internal audit of contractor management at a mine's process plant. Checklist rows converted from the procedure: “permits closed same day”, “competent person on site during hot work”, “monthly contractor review held”. Sampling: three permits across two weeks found two closed, one left open with no annotation, minor, with a root cause of the 16:30 issuing office closing before crews knock off; interviews found the competent-person role filled by whoever was available on two days, observation of a hot-work crew with no named competent person present, major, with a root cause of the appointment list being six months out of date after a reorganisation. Report: two findings, both with root causes, both with dated actions accepted at the closing meeting; the open-permit finding's fix, a 15:45 permit close-out round, cost nothing and closed the gap the next week.

7.8 Activity and knowledge check

LEARNER ACTIVITY

Convert, sample, grade

Groups receive a procedure extract and a folder of “evidence”, records seeded with two planted defects. Convert five procedure lines into evidence questions, sample the folder, write the findings in the three-sentence form with gradings, and present the report to the “auditee” group, who defend their root causes. Debrief: which planted

defect stayed hidden, and what sampling choice would have caught it?

KNOWLEDGE CHECK

Auditing and inspection techniques — check your understanding

1. Distinguish inspection, audit and verification.
2. Convert this requirement into an evidence question: “The organisation shall maintain training records.”
3. What is triangulation in auditing?
4. Define the four finding grades.
5. Why is courtesy an auditor's discipline?

7.9 Module summary

- Convert requirements into evidence questions before auditing.
- Triangulate documents, interviews and observation; sample the edges.
- Findings: requirement, evidence, gap, graded by risk, actioned to closure.
- Courtesy keeps root causes honest.

Module 8 Training of trainers

Contact time: One day · *Oryx Safety Academy*

8.1 What this module covers

The programme's final module multiplies it: how to design and deliver safety training that sticks, and how to feed new instructors back into Levels 1 and 2. It covers adult-learning principles, session design, delivery with difficult rooms, and assessment of learning, closing with each learner delivering and receiving feedback on a training segment.

8.2 Learning outcomes

By the end of this module you should be able to:

- Apply adult-learning principles: relevance, experience, practice, respect.
- Design a training session from outcomes backward: objectives, activities, checks, timing.

- Deliver a training segment managing nerves, questions and difficult participants.
- Assess learning with checks tied to the objectives, and give usable feedback.
- Describe the path to becoming an Oryx Academy assistant instructor, subject to the credential requirements.

8.3 Key terminology

Term	Working definition
Backward design	Writing objectives first, then assessments, then activities, so everything serves the outcome.
Formative check	A low-stakes question or task during training that reveals whether learning is landing.
Difficult room	Resistance patterns, silence, derailing, cynicism, and their respectful handling.
Presenter feedback	Specific, behavioural, and owned by the receiver: what worked, what to try next.

8.4 Adults learn what solves tomorrow

Adults allocate attention to what visibly solves tomorrow's problem, which is why every Oryx module opens with a workplace problem rather than a definition. Experience is an asset to harvest, not an ego to overcome: the oldest hand's story teaches better than the slide behind it. Practice beats explanation in any skill domain, and respect is the price of everything: condescension flips a room off for the day. These four, relevance, experience, practice, respect, are the design tests every session must pass.

8.5 Designing backward

Write the objectives as observable behaviours, “each learner will write a four-part hazard report”, not “understand reporting”. Design the check that would prove it. Then build the shortest path of content and practice to that check, and time it honestly, pad nothing, and cut before you pad. The session plan carries timings, materials, and the formative checks where you will discover whether to speed up or slow down.

8.6 Delivery and the difficult room

Nerves are managed by rehearsal and by knowing your first three sentences cold. Questions are honoured, and answered or owned, “I will find out and send it by Friday”, and then sent by Friday, because trainers' credibility compounds the same way sites' does. The difficult room patterns and their counters: silence, small safe questions and visible action; derailing, acknowledge, park on the flip chart, return if time; cynicism, facts, respect, and one convert at a time, never a confrontation. The trainer's face is the course's weather; keep it kind.

8.7 Becoming an Oryx instructor

The Academy's instructor path is explicit: assist on a Level 1 cohort; co-deliver under the lead instructor; deliver independently once the lead instructor signs off the observed delivery and the safety credentials are verified, the same Saiosh or SAMTRAC-class requirement the plan imposes on all lead instructors. Training of trainers here certifies teaching craft; it does not substitute for the safety credential, and it says so on the certificate.

8.7 Worked example

Worked example: A mine's training officer rebuilds the site induction using backward design. Objective: every new worker walks the site's three highest hazards and states their controls. Check: at induction's end, each worker leads a two-minute hazard walk for the group at one real location. Design: forty minutes of content, three location walks, the check, and a feedback round. The old induction, ninety minutes of slides, retired. Six months later, new workers' near-miss reporting rate in their first month, previously near zero, matched the site average, because the first day had taught them the site was interested in what they noticed.

8.8 Activity and knowledge check

LEARNER ACTIVITY

The assessed training segment

Each learner delivers a fifteen-minute training segment on a safety topic of their choice, designed backward from one written objective with one formative check. Peers assess against the course rubric: objective clarity, design economy, delivery, check effectiveness, feedback given to others. Deliveries are scheduled across the day; every learner gives and receives structured feedback, the feedback exchange being the assessed half of the exercise.

KNOWLEDGE CHECK

Training of trainers — check your understanding

1. Name the four adult-learning design tests.
2. What does backward design sequence?
3. A question you cannot answer should be…
   A. Deflected with a joke
   B. Owned with a commitment date, then met
   C. Guessed at confidently
4. Give the counter-pattern for a cynical room.
5. What does the training-of-trainers certificate mean, and not mean, on the instructor path?

8.9 Module summary

- Adults learn what solves tomorrow; harvest their experience.
- Design backward: objectives, checks, shortest path.
- Own questions with dates; manage rooms without confrontation.
- Instructor path: assist, co-deliver, observed sign-off, credentials verified.

Assignments and progression

Two assignments run across the programme: the management-system assignment set in Module 1 and the practical audit report set in Module 7. Briefs and rubrics are in the assessment pack; submissions are due three weeks after the final module day. Module examinations are sat at the end of each module block.

Graduates progressing further: the Academy's instructor path, assist, co-deliver, observed sign-off with verified credentials, is open to graduates who complete the Training-of-Trainers module and hold the required safety credential. It is a paid, contracted role, and it is how this programme intends to grow Namibian instructors rather than import them.

References and further reading

References follow a consistent author–date style; legal references cite jurisdiction, short title, year and authority. Every entry is carried in the Oryx

Safety Academy Research and Reference Register with reliability tier and review status. Where a provision's current numbering or applicability is uncertain, verify against the official gazetted text or the responsible authority before relying on it.

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[R7] International Labour Organization (1981). Occupational Safety and Health Convention (No. 155); and ILO codes of practice including Safety and Health in Construction (1992). Geneva: ILO.

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Glossary

Terms are defined as used in this manual. Where a term has a statutory definition, the statute prevails.

Term	Definition
Active failure	The triggering human act or omission in an incident.
Audit	Systematic, documented verification against a standard.
Backward design	Objectives first, then checks, then the shortest teaching path.
Code of practice (mine)	Site-specific documented procedure for a major mining hazard.
Client coordination duty	The client's non-delegable duty to manage contractor interfaces.
Gap analysis	Structured comparison of practice against a standard, prioritised.
ICAM	Incident Cause Analysis Method, organised around organisational, task/environmental, individual and precondition factors.
Latent condition	A system weakness planted before the incident that makes it possible.
Method statement	Task-specific safe system of work: sequence, hazards, controls, competencies, hold points.
Objective evidence	Re-verifiable information: records, observations, demonstrable facts.
OEL	Occupational exposure limit; source-dependent airborne concentration benchmark.
PDCA	Plan-Do-Check-Act, the management-system cycle.
Personal sampling	Measurement in a worker's breathing zone over a representative shift.
Prequalification	Evidence-based screening of contractor safety capability before award.
Prevention through design	Removing hazards on the drawing board.
SHEQ	Safety, health, environment and quality as one system.
TWA	Time-weighted average exposure over a workday.

Feedback and support

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